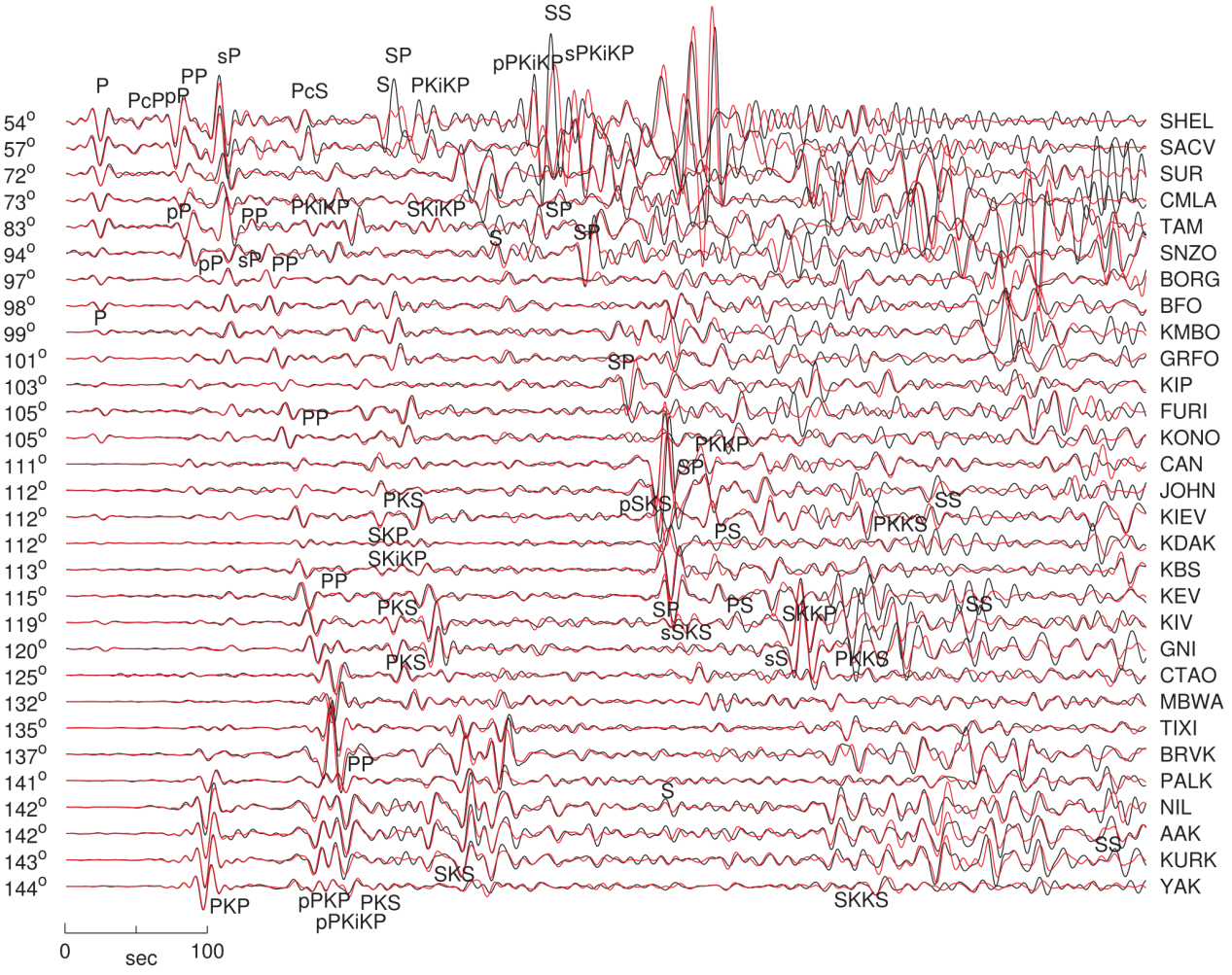
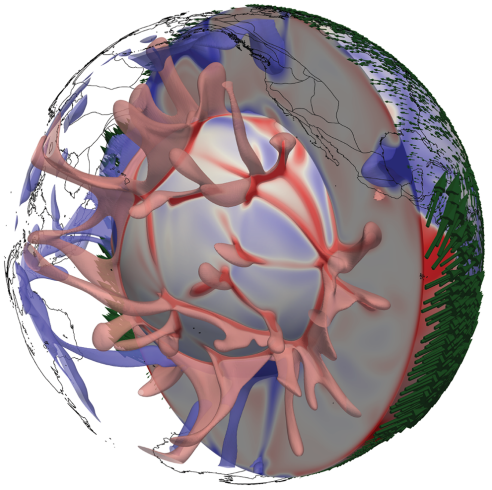
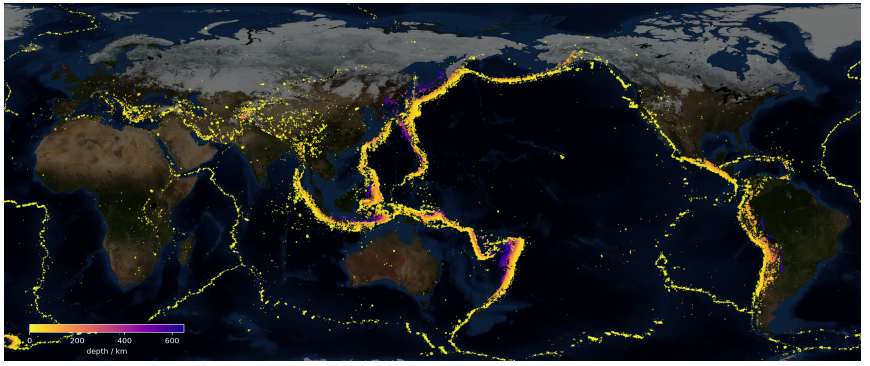
Lecture 13

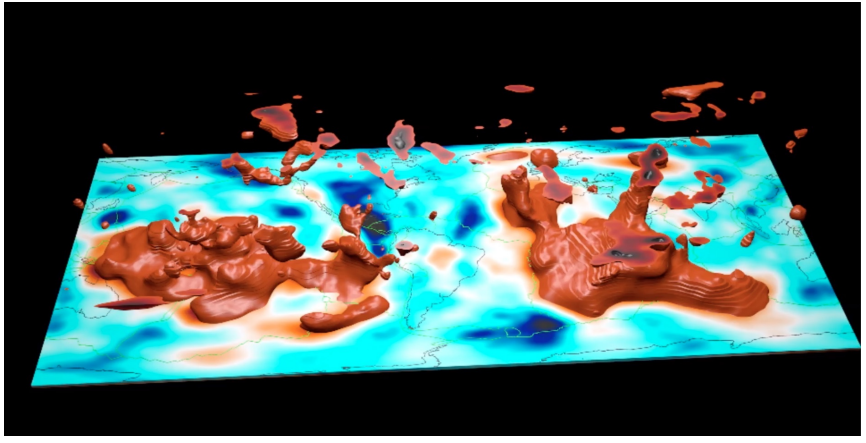
Finite elasticity

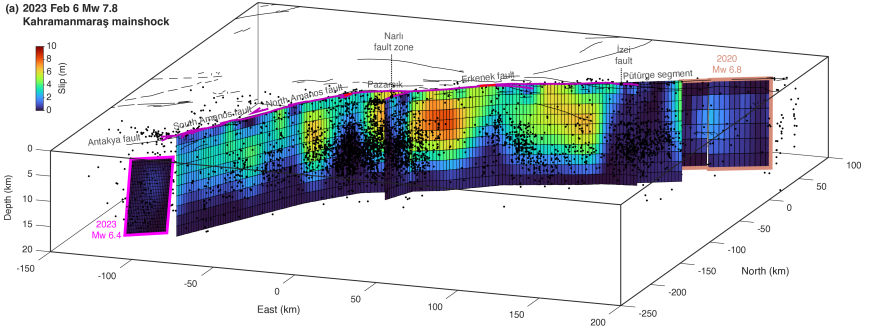


Tromp *et al.* (2010)



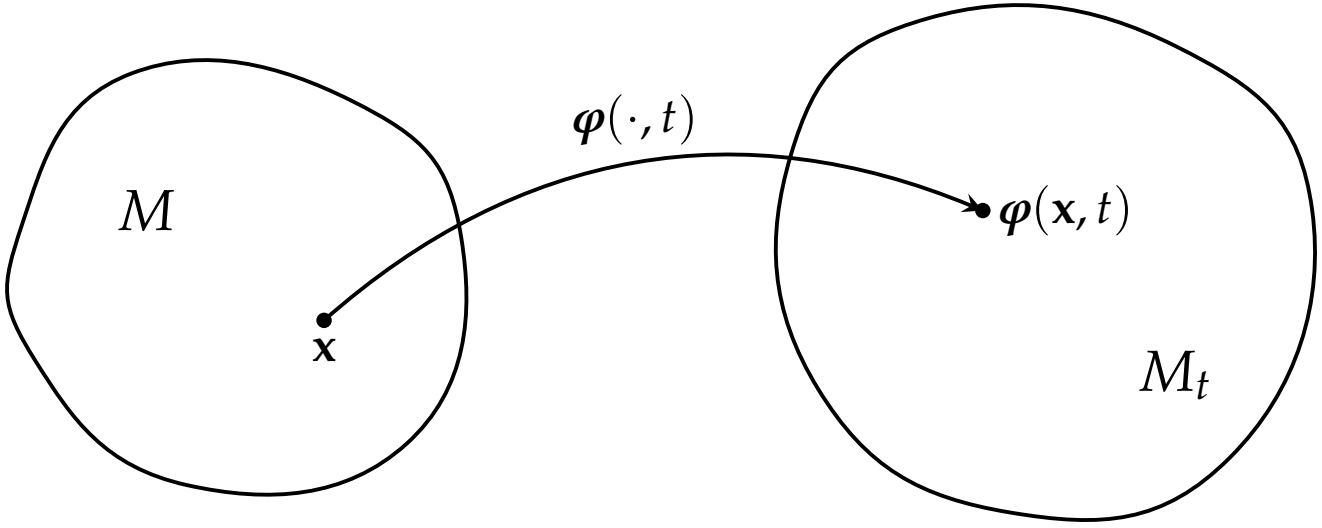






Davies *et al.* (2022); Pugmire (2025); USGS (2025); Barbot *et al.* (2023)

Kinematics



Deformation of a continuum defined relative to a reference body.

The **motion** is a field that specifies the position of particle $\mathbf{x}$ at time t in physical space.

The motion is assumed to be smooth and have a smooth inverse – not fracturing or tearing or interpenetration.

The conventional choice of reference body is based on a reference time:

$$\boldsymbol{\varphi}(\mathbf{x}, t_0) = \mathbf{x}.$$

The (referential) **velocity** is the time derivative of the motion:

$$v_i(\mathbf{x}, t) = \frac{\partial \varphi_i}{\partial t}(\mathbf{x}, t).$$

The **deformation gradient** is the spatial derivative of the motion:

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j},$$

$$\varphi_i(\mathbf{x} + \delta\mathbf{x}, t) = \varphi_i(\mathbf{x}, t) + F_{ij}(\mathbf{x}, t)\delta x_j + \cdots .$$

Because the motion maps the reference body one-to-one with the instantaneous body, the deformation gradient is at each point an invertible matrix. We are free to assume its determinant is **everywhere positive**.

The **Jacobian** of the motion is defined by:

$$J(\mathbf{x}, t) = \det \mathbf{F}(\mathbf{x}, t).$$

We can use it to change variables within volume integrals:

$$I = \int_{M_t} f(\mathbf{y}) \, \mathrm{d}^3\mathbf{y},$$

$$I = \int_{M} f[\boldsymbol{\varphi}(\mathbf{x}, t)] J(\mathbf{x}, t) \, \mathrm{d}^3\mathbf{x}.$$

In this manner, we remove the need to explicitly consider moving boundaries.

Mass of a sub-body does not change with time:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{U_t}\varrho(\mathbf{y},t)\,\mathrm{d}^3\mathbf{y}=0.$$

Transforming this to the reference body, we get:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{U}J(\mathbf{x},t)\varrho[\boldsymbol{\varphi}(\mathbf{x},t),t]\,\mathrm{d}^3\mathbf{x}=0.$$

The reference volume is arbitrary, and so the integrand does not depend on time:

$$\rho(\mathbf{x})=J(\mathbf{x},t)\varrho[\boldsymbol{\varphi}(\mathbf{x},t),t].$$

The **referential density** equals the density at the reference time.

Dynamics

Kinetic energy density per unit volume in the reference body:

$$\frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}(\mathbf{x},t)\|^2.$$

Total kinetic energy associated with a motion:

$$\mathcal{T}(t) = \frac{1}{2}\int_M \rho(\mathbf{x})\|\mathbf{v}(\mathbf{x},t)\|^2 \,\mathrm{d}^3\mathbf{x}.$$

This is just generalising particle mechanics to a continuum. What about the potential energy?

As a starting point, we assume that the potential energy can be written:

$$\mathcal{V}(t) = \int_M W\left[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)\right] \mathrm{d}^3\mathbf{x}.$$

This choice defines a **simple hyperelastic material**:

- Hyperelastic – a potential energy for the body exists (cf. conservative forces in particle mechanics)
- Simple – the potential energy density depends only on first spatial derivatives of the motion

The key physical intuition is that elastic materials have a preferred shape or volume, and that energy is required to move them into a different configuration.

Material frame indifference requires the potential energy be invariant under a set of transformations:

Time-translation:

$$\boldsymbol{\varphi}'(\mathbf{x},t) = \boldsymbol{\varphi}(\mathbf{x},t-\tau) \quad \longrightarrow \quad \mathcal{V}'(t) = \mathcal{V}(t-\tau),$$

$$\mathcal{V}(t) = \int_M W\big[\mathbf{x},\boldsymbol{\varphi}(\mathbf{x},t),\mathbf{F}(\mathbf{x},t)\big]\,\mathrm{d}^3\mathbf{x}.$$

Rigid-body motions:

$$\boldsymbol{\varphi}'(\mathbf{x},t) = \mathbf{a}(t) + \mathbf{Q}(t)\boldsymbol{\varphi}(\mathbf{x},t) \quad \longrightarrow \quad \mathcal{V}'(t) = \mathcal{V}(t),$$

$$W\big[\mathbf{x},\mathbf{F}(\mathbf{x},t)\big] = W\big[\mathbf{x},\mathbf{Q}(t)\mathbf{F}(\mathbf{x},t)\big].$$

Invariance under superimposed rigid rotations discussed further next time.

The **Lagrangian density** for a simple hyperelastic body is defined by:

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F}).$$

The corresponding **action** is given by:

$$\mathcal{S}(\boldsymbol{\varphi}) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t.$$

Hamilton's principle states that the true motion is a **stationary point** of the action for variations subject to fixed end-point conditions. This means that the **first variation of the action**

$$\delta\mathcal{S} = \lim_{h \to 0} \frac{1}{h}[\mathcal{S}(\boldsymbol{\varphi} + h\,\delta\boldsymbol{\varphi}) - \mathcal{S}(\boldsymbol{\varphi})]$$

is zero for all perturbations to the motion such that:

$$\delta\boldsymbol{\varphi}(\mathbf{x}, 0) = \delta\boldsymbol{\varphi}(\mathbf{x}, T) = \mathbf{0}.$$

From Hamilton's principle, we arrive at the **Euler–Lagrange equations**:

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial v_i}\right) - \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial F_{ij}}\right) = 0,$$

along with the **natural boundary conditions**:

$$\frac{\partial \mathcal{L}}{\partial F_{ij}}\hat{n}_j = 0.$$

Full details of the derivation are given in the handout.

Using the concrete form for the Lagrangian density, the equations of motion are:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

along with the traction-free boundary conditions:

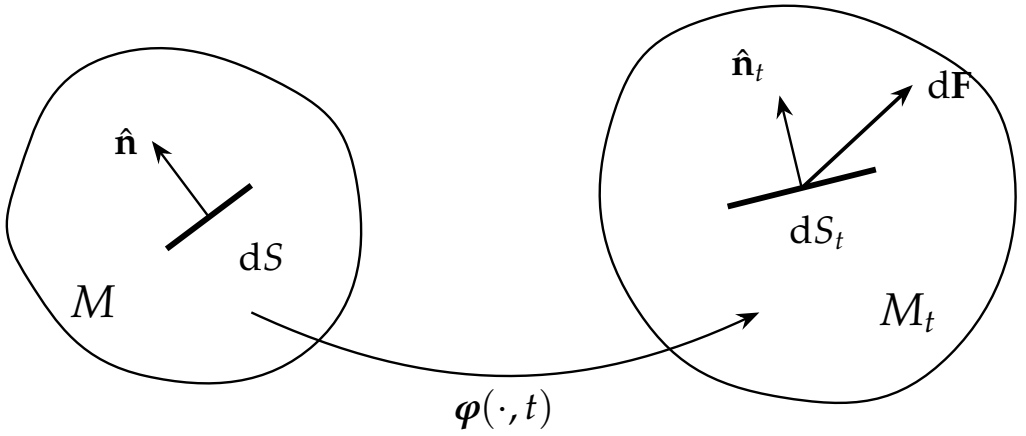
$$T_{ij} \hat{n}_j = 0,$$

where we have defined the **first Piola–Kirchhoff stress** by:

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)].$$

This tensor is the referential counterpart of the **Cauchy stress** seen within spatial (Eulerian) formulations of continuum mechanics.

In a referential (Lagrangian) framework, conservation of mass is trivially met, while the equations of motion are posed on a fixed reference body.



$$\mathrm{d}\mathbf{F} = \mathbf{T}\hat{\mathbf{n}}\,\mathrm{d}S = \sigma\hat{\mathbf{n}}_t\,\mathrm{d}S_t.$$

The first Piola–Kirchhoff stress, $\mathbf{T}$, gives the force on the deformed surface element per unit area of the reference element; the Cauchy stress, σ, gives the same force per unit area of the deformed element.

What you need to know and be able to do

(i) Understand the terminology needed to describe the kinematics of an elastic body.

(ii) Use the conservation of mass to derive the relation between the spatial and referential densities.

(iii) State expressions for the kinetic and potential energies of a simple hyperelastic body, and explain the underlying physical ideas and assumptions.

(iv) Give a statement of Hamilton’s principle in the context of finite elasticity, and use it to derive the equations of motion and associated boundary conditions.